

Exercise 6: MLP Optimization on Breast Cancer Dataset

Follow class_exercise_5 to complete the following tasks. You should write code in a notebook and upload in the moodle.

Task 1: Load and Prepare a New Dataset

Instead of Iris or Wine, you will use the **Breast Cancer Wisconsin (Diagnostic)** dataset.

1. Load the dataset using `sklearn.datasets.load_breast_cancer()`.
 2. Split the data into **Train (60%)**, **Validation (20%)**, and **Test (20%)** sets. Use `SEED = 42` for reproducibility.
 3. Standardize the features using the `standardize_train_test` logic from your previous notebook (calculate mean/std on the training set only and apply to val/test).
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Task 2: Building the Hybrid Activation MLP

Construct a Multi-Layer Perceptron (MLP) using either **PyTorch** or **TensorFlow** with the following specific architecture:

- **Input Layer:** Matching the number of features in the Breast Cancer dataset (30 features).
 - **Hidden Layer 1:** 64 units, **ReLU** activation.
 - **Hidden Layer 2:** 32 units, **Tanh** activation.
 - **Hidden Layer 3:** 16 units, **Sigmoid** activation.
 - **Output Layer:** Since this is a binary classification task (Malignant vs. Benign), use 1 unit with a Sigmoid activation (or 2 units with Softmax).
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Task 3: Optimization Sweep

Experiment with different optimization algorithms to see how they impact the learning process. Keep the learning rate constant at `0.001` and train for `100 epochs`.

Compare the following three optimizers:

1. **SGD** (Stochastic Gradient Descent)
2. **RMSprop**
3. **Adam**

Task:

For each optimizer:

- Train the model.
 - Record the **Training Loss** and **Validation Accuracy** at each epoch.
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Exercise 4: Visualization and Analysis

Create two side-by-side plots to compare the performance of the three optimizers:

1. **Plot A (Loss Curves):** Plot the training loss vs. epochs for SGD, RMSprop, and Adam on the same graph.
2. **Plot B (Accuracy Curves):** Plot the validation accuracy vs. epochs for all three optimizers on the same graph.

Discussion Question: Based on your plots, which optimizer reached a stable accuracy the fastest? Did any optimizer show signs of oscillation or slow convergence? Document your findings below the plots.

After